

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I, M. Mary Shalin (192511330) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Mary Shalin

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CLOUD COMPUTING FUNDAMENTALS

CLOUD COMPUTING

A model for enabling ubiquitous, convenient, on demand network access to a shared pool of configurable computing resources

Key characteristics

- On demand access
- Broad network access
- Rapid elasticity
- Resource Pooling

Benefits

- Cost effectiveness
- Scalability
- High availability
- Flexibility
- Global access

Limitations

- Security issues
- Privacy
- Internet dependency
- Downtime
- Vendor lock

DELIVERS VALUE TO

Providers

- Better resource utilization
- Reduced operational cost
- Wider reach

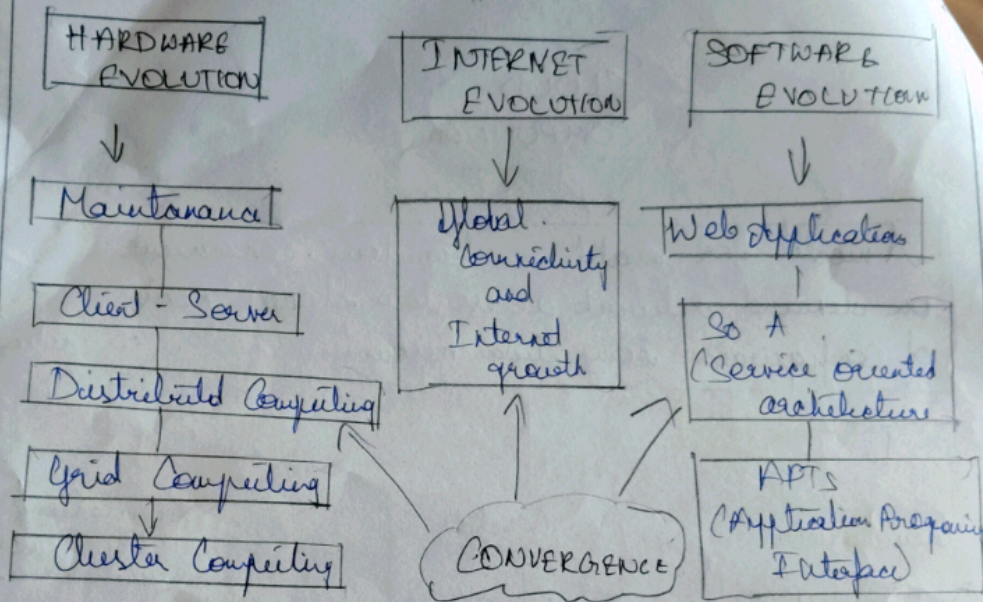
Through

Consumers

- Pay as you go
- Focus on core business
- Access to latest technology

2).

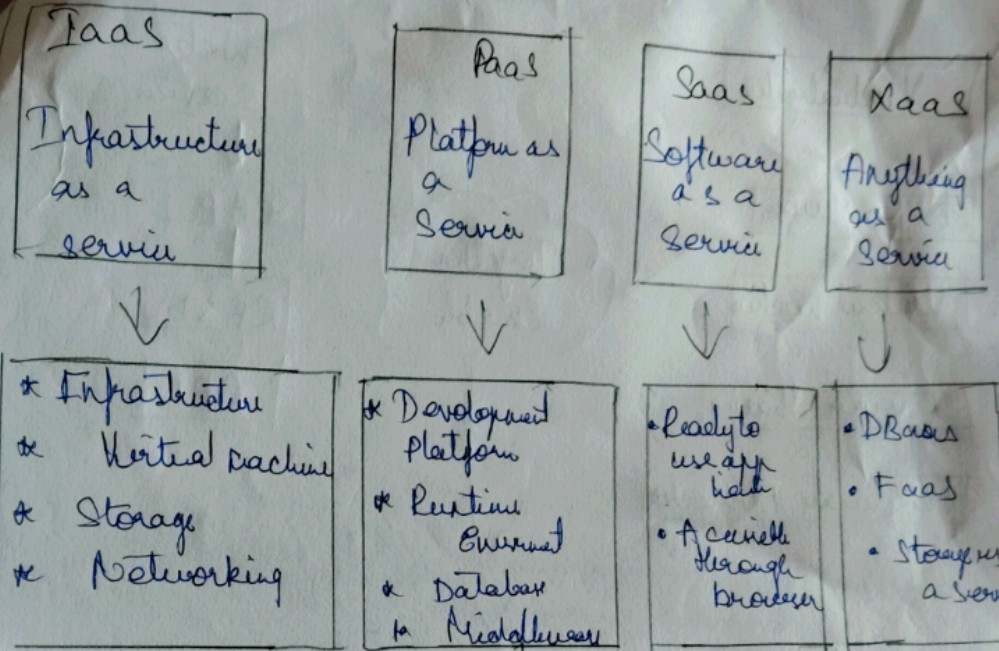
EVALUATION OF CLOUD COMPUTING



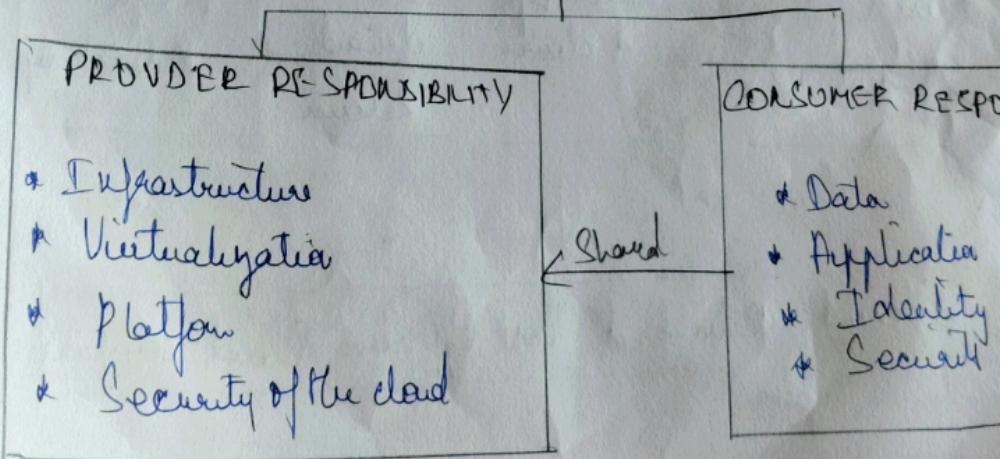
SERVICE DELIVERY EVOLUTION



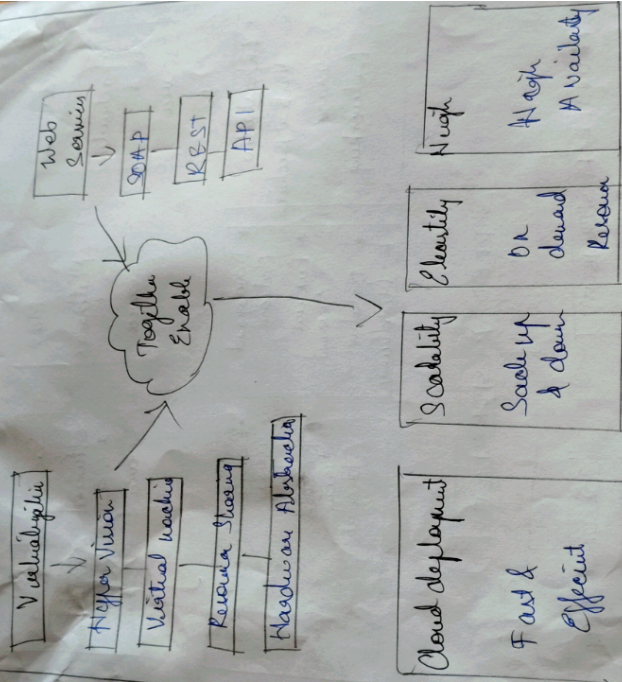
Cloud Service Models



SHARED RESPONSIBILITY



6) VIRTUALIZATION AND WEB SERVICE:



Efficient, Reliable & Cost effective Cloud computing